Problem 1. Let $a \in \mathbb{R}$. Prove that if $0 < a < 1$, then $0 < a^2 < a$. (Notice that $0 < a < 1$ means that both $0 < a$ and $a < 1$.)

You can *only* use the axioms on the "Axioms" handout and results that we have proved that appear on that document.

Problem 2. Let $x \in \mathbb{R}$. The **absolute value** of x is defined by

$$|x| = \begin{cases} x & \text{if } x \geqslant 0 \\ -x & \text{if } x < 0 \end{cases}.$$

Show for all $x, y \in \mathbb{R}$ that $|xy| = |x| \cdot |y|$.

Problem 3. In this problem, you will show that $\sqrt{2} + \sqrt{3}$ is irrational.

1. Prove that if $x \in \mathbb{R}$ is irrational and $m \in \mathbb{Q}\backslash\{0\}$, then xm is irrational.
2. Prove that $\sqrt{6}$ is irrational.
3. Prove that $\sqrt{2} + \sqrt{3}$ is irrational.

Problem 4. For each of the following proposed theorems, provide a proof if the theorem is true, or provide a counterexample if it is false.

1. **Theorem (?)**

   Let $a, b \in \mathbb{Z}$. If $ab \equiv 0 \pmod 3$, then either $a \equiv 0 \pmod 3$ or $b \equiv 0 \pmod 3$.

2. **Theorem (?)**

   Let $a, b \in \mathbb{Z}$. If $ab \equiv 0 \pmod 4$, then either $a \equiv 0 \pmod 4$ or $b \equiv 0 \pmod 4$.

Problem 5. For which natural numbers is $n! > 3^n$? Prove it.

Problem 6. Is the following proposition true or false? Prove or give a counterexample.
For each nonnegative integer n, $8^n \mid (4n)!$.

Problem 7. 1. For $n \in \{1, 2, \ldots, 7\}$, fill out the following table.

n	$1^3 + 2^3 + \cdots + n^3$	$n(n+1)$
1	1	2
	etc.	

2. Conjecture an equality relating $1^3 + 2^3 + \cdots + n^3$ and $n(n+1)$.
3. Prove that your conjecture is correct.

Problem 8. Prove that the sum of the cubes of any three consecutive natural numbers is divisible by 9. (For example, $4^3 + 5^3 + 6^3 = 405 = 9 \cdot 45$.)

Problem 9. Complete Exercise 18 (a) and (b) on page 184 of the textbook (*Mathematical Reasoning* by Sundstrom).

Problem 10. Do not turn these in, but I recommend doing the progress checks for sections 3.4 and 4.1 in your textbook.